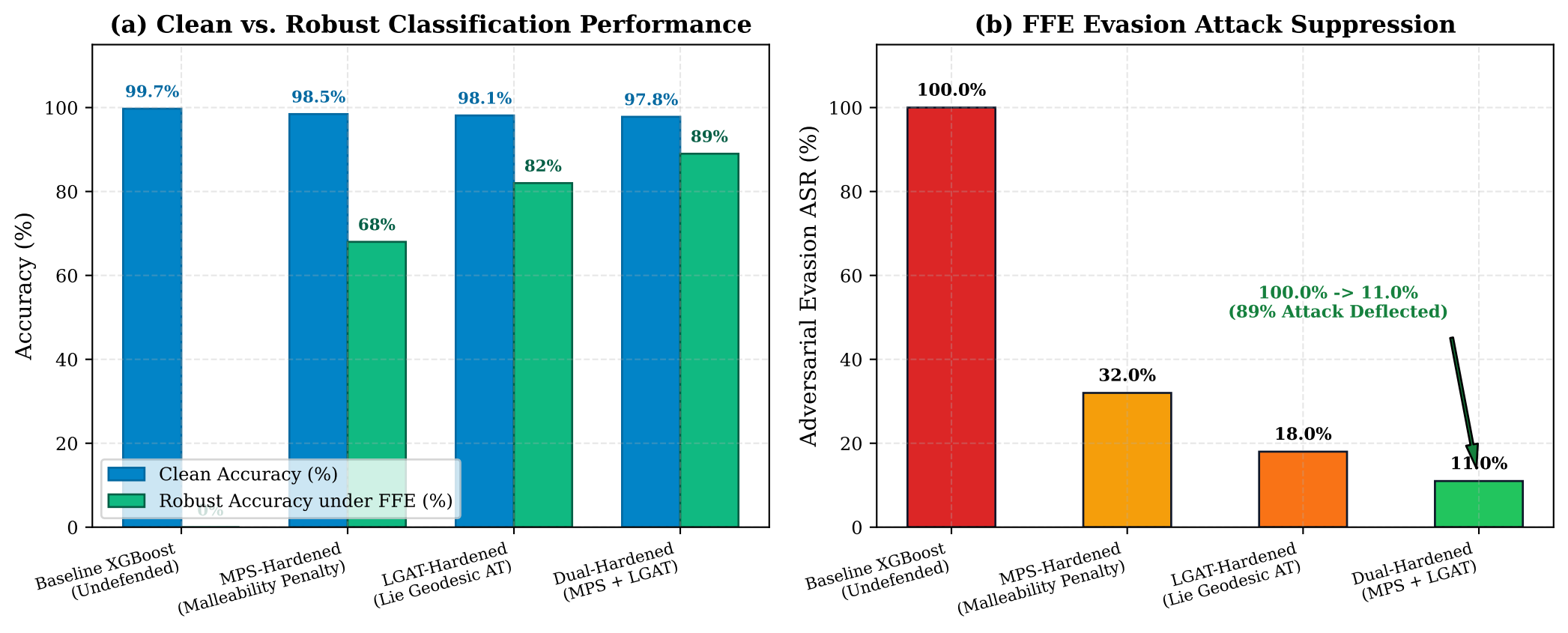
(a) Clean vs. Robust Classification Performance
Accuracy (%)
99.7%
98.5%
98.1%
97.8%
68%
82%
89%
Clean Accuracy (%)
Robust Accuracy under FFE (%)
Baseline XGBoost (Undefended)
MPS-Hardened (Malleability Penalty)
LGAT-Hardened (Lie Geodesic AT)
Dual-Hardened (MPS + LGAT)
(b) FFE Evasion Attack Suppression
Adversarial Evasion ASR (%)
100.0%
32.0%
18.0%
11.0%
100.0% -> 11.0%
(89% Attack Deflected)